

2017 Maths Entrance Exam

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) Mathematics for Natural, Sene, 2017 E.C./June/July, 2025 G.C.

NUMBER OF ITEMS: 60
TIME ALLOWED: 3 HOURS

SUBJECT CODE: 02
BOOKLET CODE: 291



EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES)
ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE
EXAMINATION (ESSLCE)
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- Which one of the following is an irrational number?
A. $\sqrt{0.04}$
B. 1.16222...
C. $\sqrt{29}$
D. 3.4000
- Which of the following is the simplified form of the expression $\frac{81^{\frac{1}{5}}}{27^{\frac{3}{5}}}$?
A. $\sqrt[5]{9}$
B. $\sqrt[5]{3}$
C. $\frac{1}{3}$
D. 3
- When the number $43 \times 10^8 \times 10^{-5}$ is written in standard form, it equals:
A. 43×10^3
B. 4.3×10^3
C. 0.43×10^5
D. 4.3×10^4
- Given the system of linear equations $\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$, where $a_1, a_2, b_1, b_2, c_1, c_2 \in \mathbb{R}$ and $a_2, b_2, c_2 \neq 0$, then which of the following is true?
A. If $\frac{a_1}{a_2} = \frac{c_1}{c_2}$, then the system has no solution.
B. If $\frac{b_1}{b_2} = \frac{c_1}{c_2}$, then the system has unique solution.
C. If $\frac{a_1}{a_2} \neq \frac{c_1}{c_2}$, then the system has infinitely many solutions.
D. If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the system has one solution.

5. If one of the roots of the equation $2x^2 - 8x + k = 0$ exceeds the other by 2, then what is the value of k ?
- A. 6
B. 4
C. 3
D. 1
- Handwritten solution:*
 $2(2)^2 - 8(2) + k = 0$
 $8 - 16 + k = 0$
 $-16 + k = 0$
 $k = 16$

6. Let $A = \{x \in \mathbb{Z} : -2 \leq x \leq 2\}$ and $B = \{x \in \mathbb{N} : 2x \leq 3x + 1\}$ are given two sets. Then what is $n(A \cup B)$?
- A. 2
B. 6
C. 5
D. 10

7. Let $A = \{x | 1 \leq x \leq 10\}$ and $B = \{x | -1 \leq x \leq 10\}$ be two sets of real numbers. What is the relative complement of A with respect to B ?
- A. ϕ
B. $\{x | -1 \leq x \leq 1\}$
C. $\{x | -1 \leq x < 1\}$
D. $\{x | -1 \leq x \leq 0\}$

Which of the following statement is true about a relation?

- A. The domain and the range of a relation are equal sets.
 B. The range of a relation is the set of all its first coordinates.
 C. The domain of a relation is the set of all its second coordinates.
 D. A relation from A to B is any subset of $A \times B$ for non-empty sets A and B .

What is the minimum value of the function defined by $f(x) = -3 + 4x + x^2$?

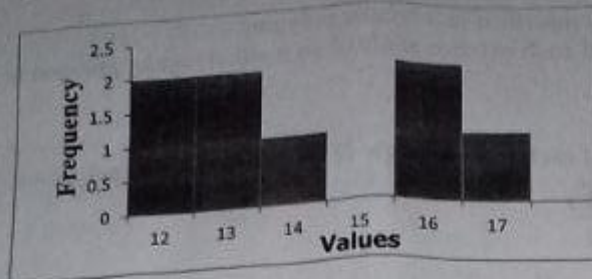
- A. 1
B. 0
C. -2
D. -7

10. Which of the following is **NOT** true about regular polygons?
- A. It can be inscribed in a circle.
 - B. A circle can be inscribed in a regular polygon.
 - C. The measure of each exterior angle of an n -sided regular polygon is equal to $\frac{360^\circ}{n}$.
 - D. The measure of each interior angle of an n -sided regular polygon is equal to $\frac{n(180^\circ)}{n-2}$.
11. Suppose two triangles are similar by SSS similarity theorem. If the side lengths of one triangle are 2cm, 4cm and 6cm, and the shorter side of the other triangle is 9cm long, then what are the lengths (in cm) of the other two sides of the triangle?
- A. 18 and 27
 - B. 9 and 27
 - C. 18 and 21
 - D. 9 and 21
12. A boy and girl are both standing on a sun light. The boy's height is 1.6m and casts a shadow of 2m. If the girl's height casts a shadow of 1.8m, then what is the height (in meter) of the girl?
- A. 1.54
 - B. 1.68
 - C. 1.44
 - D. 1.5

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13. Consider the following histogram.



Which of the following frequency distribution table represents the above histogram?

A.

V	12	13	14	15	16	17
f	2	1	1	0	2	1

B.

V	12	13	14	15	16	17
f	2	1	1	1	2	1

C.

V	12	13	14	15	16	17
f	2	2	1	0	2	1

D.

V	12	13	14	15	16	17
f	2	2	1	2	2	1

14. If a fair six sided die is tossed once, then what is the probability of getting a factor of 4?

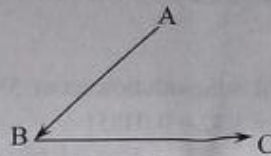
A. $\frac{1}{6}$

C. $\frac{1}{3}$

B. $\frac{5}{6}$

D. $\frac{1}{2}$

15. If two vectors $\vec{V} = \vec{AB}$, and $\vec{U} = \vec{BC}$ are given as shown in the figure below.



Then what is the vector $\vec{V} + \vec{U}$?

A. $\vec{BA}$

C. $\vec{CB}$

B. $\vec{CA}$

D. $\vec{AC}$

16. Let $f(x) = x^3 + 2x + 1$ and $g(x) = x^2 - 1$ be given two polynomial functions. Then which of the following is true?

A. $f(x) + g(x) = x^3 + x^2$

B. $f(x) - g(x) = x^3 - x^2 + 2x$

C. $f(x) \cdot g(x) = x^5 + x^3 + x^2 - 2x - 1$

D. $\frac{f(x)}{g(x)} = \frac{x^2 + 1}{x - 1}$

17. Let $a, b \in \mathbb{R}$ such that $a < b$ and f be a polynomial function. Then which one of the following statement states the location theorem?
- A. If $f(a)$ and $f(b)$ have the same signs, then there is at least one zero of f between a and b .
 - B. If $f(a)$ and $f(b)$ have the same signs, then there exist a unique zeros of f between a and b .
 - C. If $f(a)$ and $f(b)$ have opposite signs, then there is a unique zeros of f between a and b .
 - D. If $f(a)$ and $f(b)$ have opposite signs, then there is at least one zero of f between a and b .
18. Which one of the following is the solution set of $5^x = 1.12$?
[Use $\log 5 = 0.6990$ and $\log 1.12 = 0.0485$]
- A. 1.5
 - B. 1.2
 - C. 1.69
 - D. 1.12
19. Given that $\log 4.8 = 0.6812$, $\log 6.93 = 0.8406$ and $\log 6.84 = 0.8351$. Then what is the value of $\sqrt{48}$?
- A. 6.71
 - B. 6.93
 - C. 6.84
 - D. 6.83
20. Consider the equation $\log x + \log (x-3) = 1$. Then what is its solution set?
- A. $\{2, -5\}$
 - B. $\{-2, 5\}$
 - C. $\{6\}$
 - D. $\{5\}$
1. If the number of bacteria in a certain laboratory colony grows at the rate of 5% per day and assuming that there are 1000 bacteria present initially, then what would be the approximate number of bacteria present after 3 days?
- A. 1102
 - B. 1158
 - C. 1050
 - D. 1105

- A. (2, 0) C. (3, 0)
B. (2, 1) D. (3, 1)

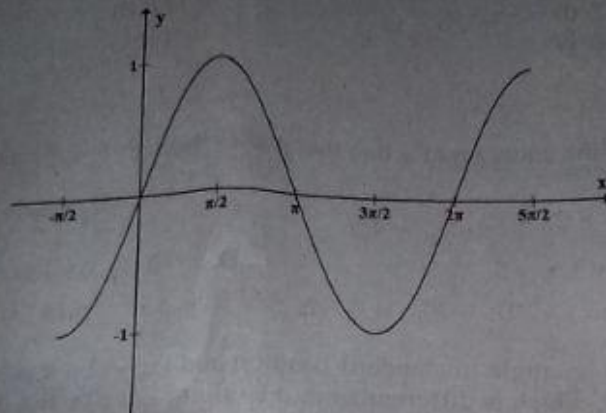
23. What is the equation of a line that passes through $(-2, 5)$ and has a slope $\frac{1}{2}$?

- A. $y = 2x + 12$ C. $y = \frac{1}{2}x + 5$
B. $2y + x = 5$ D. $2y - x - 12 = 0$

24. Let θ be an angle in standard position and (x, y) be a point on the terminal side of θ , which is different from the origin. Let r be the distance from (x, y) to the origin. Then which of the following defines $\cos \theta$?

- A. $\frac{y}{r}$
- B. $\frac{r}{x}$
- C. $\frac{x}{r}$
- D. $\frac{r}{y}$

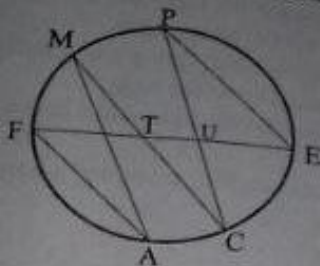
25. Consider the graph of a certain trigonometric function given below.



Which one of the following functions represents the above graph for

$$-\frac{\pi}{2} \leq x \leq \frac{5\pi}{2}?$$

- | | |
|-----------------|-----------------|
| A. $y = \sin x$ | C. $y = \cos x$ |
| B. $y = \csc x$ | D. $y = \sec x$ |
26. Which of the following is true about the cosecant value of the given angles?
- | | |
|---|---|
| A. $\csc 135^\circ = -\sqrt{2}$ | C. $\csc 45^\circ = \sqrt{2}$ |
| B. $\csc 45^\circ = \frac{1}{\sqrt{2}}$ | D. $\csc 135^\circ = -\frac{1}{\sqrt{2}}$ |



A. 30°

C. 25°

B. 50°

D. 60° A. 28π C. 98π B. 36π

D. 49π

A. $3\sqrt{2}$

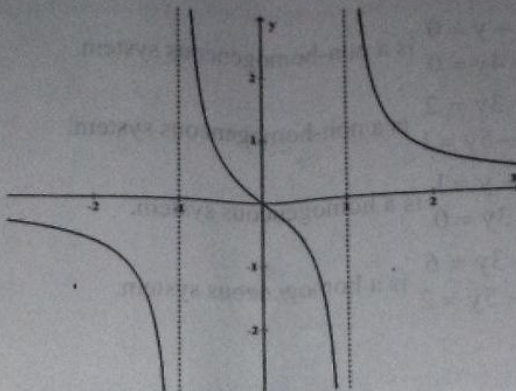
C. $36\sqrt{2}$

B. 48

D. 72

30. Which of the following statement defines a frustum of a cone?
- It is a part of a cone included between the base and a horizontal cross-section made by a plane parallel to the base.
 - It is a part of a cone included between the vertex and a horizontal cross-section made by a plane parallel to the base.
 - It is a part of a cone included between the base and a horizontal cross-section made by a plane.
 - It is a part of a cone included between the vertex and a horizontal cross-section made by a plane.
31. Which one of the following function is **NOT** a power function?
- $f(x) = \sqrt{2} x^{\frac{3}{2}}$
 - $f(x) = \pi x^5$
 - $f(x) = (\sqrt{3})^x$
 - $f(x) = x^{-\frac{3}{5}}$
32. Which of the following statement define the greatest integer function given by $f(x) = \lfloor x \rfloor$?
- It is an integer less than or greater than x .
 - It is an integer less than or equal to x .
 - It is the least integer greater than x .
 - It is an integer greater than or equal to x .
33. Which one of the following expression is the correct decomposition of the rational expression $\frac{2x-3}{x^2+x-2}$?
- $\frac{1}{3-3x} + \frac{7}{3x+6}$
 - $\frac{1}{1-x} + \frac{5}{3x+6}$
 - $\frac{7}{3x+2} - \frac{1}{x+1}$
 - $\frac{5}{3x+2} + \frac{1}{x+1}$

34. Consider the graph of the rational function given below.



Then which of the following rational function is represented by the above graph?

A. $f(x) = \frac{x}{x^2 - 1}$

C. $f(x) = \frac{x^2}{x^2 - 1}$

B. $f(x) = \frac{x}{x^2 + 1}$

D. $f(x) = \frac{x^2}{1 - x^2}$

35. If $\begin{pmatrix} x+y & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ 5 & x-y \end{pmatrix}$, then what is the value of x ?

A. -5

C. 2

B. 5

D. -2

36. Which of the following statement is true about the given system of linear equations?

A. $\begin{cases} 2x + y = 0 \\ x - 4y = 0 \end{cases}$ is a non-homogeneous system.

B. $\begin{cases} x + 3y = 2 \\ 2x - 5y = 1 \end{cases}$ is a non-homogeneous system.

C. $\begin{cases} 3x - y = 1 \\ x + 3y = 0 \end{cases}$ is a homogeneous system.

D. $\begin{cases} x + 3y = 6 \\ 2x - 5y = 2 \end{cases}$ is a homogeneous system.

37. What is the determinant of the matrix $\begin{pmatrix} \frac{1}{2} & 1 & \frac{3}{2} \\ 2 & \frac{5}{2} & 3 \\ \frac{7}{2} & 4 & \frac{9}{2} \end{pmatrix}$?

A. 0

B. $\frac{1}{2}$

C. $\frac{1}{8}$

D. 1

38. Which of the following matrix is the inverse of the matrix $\begin{pmatrix} 1 & -1 & 2 \\ 1 & 0 & 1 \\ 3 & 1 & 3 \end{pmatrix}$?

A. $\begin{pmatrix} 1 & 5 & -1 \\ -3 & 0 & 1 \\ -4 & -1 & 1 \end{pmatrix}$

C. $\begin{pmatrix} -1 & 0 & 1 \\ 5 & 3 & 4 \\ 1 & -1 & 1 \end{pmatrix}$

B. $\begin{pmatrix} -1 & 5 & -1 \\ 0 & -3 & 1 \\ 1 & -4 & 1 \end{pmatrix}$

D. $\begin{pmatrix} -1 & 0 & 1 \\ 5 & -3 & -4 \\ -1 & 1 & 1 \end{pmatrix}$

39. Which of the following statement describes the difference between scalar and vector quantities?

- A. Physical quantities that only have magnitude are scalars, while that have both magnitude and direction are vectors.
- B. Physical quantities that only have magnitude are vectors, while that have both magnitude and direction are scalars.
- C. Physical quantities that only have direction are vectors, while that have both magnitude and direction are scalars.
- D. Physical quantities that only have direction are scalars, while that have only magnitude are vectors.

40. Which of the following statement is true about types of data?

- A. The height of students in a class is a qualitative data.
- B. The weight of students in class is a qualitative data.
- C. The gender of students in a class is a quantitative data.
- D. The age of students in a class is a quantitative data.

41. Consider the following grouped frequency distribution table:

X	0-9	10-19	20-29	30-39	40-49
f	4	3	6	5	2

Then what is the class midpoint of the fourth class?

A. 34.5

C. 24.5

B. 35.5

D. 55.5

42. The table given below shows the ages distribution of children in a small village in years.

Age interval	Frequency(f)
2- 4	4
5- 7	1
8- 10	2
11- 13	3

Then what is the mean age of the children's (in years)?

A. 6.9

C. 7.2

B. 7.5

D. 6.6

43. Which one of the following distribution is multimodal?

A. 6, 10, 9, 6, 10, 4

B. 8, 7, 11, 13, 17, 9, 8, 7, 6, 11

C. 12, 13, 26, 15, 25, 38, 28, 27, 32

D. 18, 14, 15, 10, 11, 3, 18, 12

44. Suppose that we have 4 objects chosen 3 at a time. Then which of the following statement is true?
- A. The number of combinations is 24.
 - B. The number of permutations is 24.
 - C. The number of permutations is 6.
 - D. The number of combinations is 6.
45. Which one of the following statements defines favorable outcomes in a random experiment?
- A. The outcomes containing all sample elements.
 - B. The outcomes containing only one sample element.
 - C. The outcomes that have equal chance of being chosen.
 - D. The outcomes that insure the occurrence of a particular result.
46. Which of the following statement defines geometric progression?
- A. It is a sequence in which the difference between consecutive terms is a positive constant.
 - B. It is a sequence in which the difference between consecutive terms is a non-zero constant.
 - C. It is a sequence in which the ratio between consecutive terms is a positive constant.
 - D. It is a sequence in which the ratio between consecutive terms is a non-zero constant.
47. Given the sequence 3, 6, 12, 24,..., Then what is the 10th partial sum of the sequence?
- | | |
|---------|---------|
| A. 3075 | C. 1027 |
| B. 1023 | D. 3069 |

48. Which of the following statement is true about the given infinite series?

A. $\sum_{n=1}^{\infty} \left(\frac{4}{3}\right)^n$ is convergent.

C. $\sum_{n=1}^{\infty} (-1)^n \left(\frac{2}{5}\right)^n$ is divergent.

B. $\sum_{n=1}^{\infty} 2^n$ is divergent.

D. $\sum_{n=1}^{\infty} (-1)^n 2^n$ is convergent.

49. What is the average rate of change of $g(x) = x^3 - 2x$ over the interval $1 \leq x \leq 6$?

A. 41

C. 46

B. $\frac{37}{5}$

D. $\frac{203}{5}$

50. Suppose $f(x) = 3\sqrt{x}$ and $g(x) = \frac{1}{3}x^4 + 2x$ be given two functions. Then which of the following is equal to $(f+g)'(x)$?

A. $\frac{2}{3}x^{-1/3} + 4x^3 + 2x$

C. $\frac{3}{2}x^{-1/2} + \frac{4}{3}x^3 + 2$

B. $\frac{2}{3}x^{-1/3} + 4x^3 + 2$

D. $3x^{-1/2} + 4x^3 + 2$

51. Let $f(x) = (1-2x)^{\frac{1}{2}}$. Then what is the values of $f'(x)$?

A. $-\frac{1}{2}\sqrt{1-2x}$

C. $-\frac{1}{\sqrt{1-2x}}$

B. $\frac{1}{2}\sqrt{1-2x}$

D. $\frac{1}{\sqrt{1-2x}}$

52. Consider the function $f(x) = x^3 - 3x^2 + 3$ on $(-2, 3)$. Then which of the following point is local minimum and local maximum of $f(x)$ respectively?
- A. $(0, 3)$ and $(2, -1)$ C. $(2, -1)$ and $(0, 3)$
B. $(3, 3)$ and $(-2, -27)$ D. $(-2, -27)$ and $(0, 3)$
53. Which one of the following statements describes the dispersion of the data values?
- A. It measures the variability in terms of different units of data.
B. The degree to which numerical data tends to distribute about an average.
C. It helps to describe the degree of similarity of data values.
D. It compares the variability of two or more data that are dependent of the units of measurement.
54. Consider the following data:
13, 16, 19, 19, 19, 19, 24, 27
What is the mean deviation about the mode?
- A. 2.75 C. 3
B. 3.75 D. 4
55. What is the coefficient of quartile deviation for the following data?
29, 44, 46, 52, 35, 35, 25, 15, 10, 55, 56
- A. 0.35 C. 2.08
B. 0.48 D. 2.85

56. Consider the following data;
1, 3, 7, 9, 11, 13, 18, 28
Then the coefficient of mean deviation about the median is equal to:

A. $\frac{8}{5}$

C. $\frac{1}{2}$

B. $\frac{5}{8}$

D. $\frac{1}{4}$

57. Given the objective function $Z = -x + 2y$ that is subjected to the

$$\text{constraints } \begin{cases} x \geq 3 \\ x + y \geq 5 \\ x + 2y \leq 6 \\ y \geq 0 \end{cases}$$

What are the maximum and the minimum values of Z respectively?

A. 1 and -2

C. -6 and -2

B. -2 and -6

D. -2 and 1

58. A television company assembles flat screen and curved screen televisions. The company has to invest Birr 15,000 to assemble one flat screen and Birr 20,000 to assemble one curved screen television. The company allocates a maximum of Birr 900,000 in a week for assembling. The company gets Birr 1800 profit from a sale of one flat screen television and Birr 2000 profit from a sale of one curved screen television. Moreover, the company has a store of maximum capacity of storing 50 televisions. Let x be the number of flat screens and y be the number of curved screens. Then which one of the following inequalities should model the profit for the objective function $Z=1800x+2000y$?

- A.
$$\begin{cases} 15,000x + 20,000y \leq 900,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$$
- B.
$$\begin{cases} 15,000x + 20,000y \leq 360,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$$
- C.
$$\begin{cases} 20,000x + 15,000y \leq 900,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$$
- D.
$$\begin{cases} 20,000x + 15,000y \leq 360,000 \\ x + y \leq 50 \\ x, y \geq 0 \end{cases}$$

59. A company offers a discount of 25% on one of its product for promotion. What is a regular price of the product if it is purchased for Birr 99?

- A. 198 C. 125
B. 132 D. 110

60. A worker of a company borrowed Birr 100,000 from the company's credit and saving association. She agreed to pay 10% of the loan during the agreement and on the remaining amount to pay Birr 3000 (monthly plus interest) in 36 monthly installments. What is the total amount of money she paid to the credit and saving association?
- A. Birr 118,000 C. Birr 116,000
B. Birr 108,000 D. Birr 106,000

Answer With Explanation

1.

Answer: C

Explanation:

- A: $\sqrt{0.04} = 0.2$, which is rational (terminating decimal).
- B: $1.16222\ldots$ is a repeating decimal (rational).
- C: $\sqrt{29}$ is irrational because 29 is not a perfect square, and its decimal expansion is non-terminating and non-repeating.
- D: 3.4000 is rational (terminating decimal).

2.

Answer: D

Explanation:

- $\frac{81\%}{27\%} = \frac{81/100}{27/100} = \frac{81}{27} = 3.$
- This is a straightforward division of percentages.

3.

Answer: D

Explanation:

- Simplify: $43 \times 10^8 \times 10^{-5} = 43 \times 10^3 = 43,000.$
- Standard form requires $a \times 10^n$ where $1 \leq |a| < 10$, so $43,000 = 4.3 \times 10^4.$

4.

Answer: D

Explanation:

- For a unique solution, the condition is $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (the lines intersect at one point).
- A is false (it may or may not have a solution).
- B is false (does not guarantee a unique solution).
- C is false (it may have no solution or a unique solution).

5.

Answer: A

Explanation:

- Let the roots be r and s , with $r = s + 2.$
- Sum of roots: $r + s = \frac{8}{2} = 4 \rightarrow (s + 2) + s = 4 \rightarrow 2s = 2 \rightarrow s = 1, r = 3.$
- Product of roots: $r \cdot s = \frac{k}{2} \rightarrow 1 \cdot 3 = \frac{k}{2} \rightarrow k = 6.$

6.

Answer: C

Explanation:

- $A = \{-2, -1, 0, 1, 2\}$ (integers from -2 to 2), so $n(A) = 5$.
- B : Solve $2x \leq 3x + 1 \rightarrow -x \leq 1 \rightarrow x \geq -1$. Since $x \in \mathbb{N}$ (natural numbers, starting from 1), $B = \{1, 2, 3, \dots\}$. However, this is infinite, but the options suggest a finite set. Assuming a typo in the inequality (likely $2x \leq 3$), then $B = \{1\}$ (since $x = 1$ satisfies $2 \leq 3$, but $x = 2$ gives $4 > 3$).
- $A \cup B = \{-2, -1, 0, 1, 2\}$ (since $1 \in A$), so $n(A \cup B) = 5$.

7.

Answer: C

Explanation:

- Relative complement of A with respect to B is $B - A = \{x \in B \mid x \notin A\}$.
- $A = [1, 10]$, $B = [-1, 10]$, so $B - A = [-1, 1)$ (i.e., $-1 \leq x < 1$).

8.

Answer: D

Explanation:

- By definition, a relation from A to B is a subset of $A \times B$.
- A is false (domain and range need not be equal).
- B is false (range is the set of second coordinates).
- C is false (domain is the set of first coordinates).

9.

Answer: D

Explanation:

- Rewrite: $f(x) = x^2 + 4x - 3$.
- Minimum at $x = -\frac{b}{2a} = -\frac{4}{2} = -2$.
- $f(-2) = (-2)^2 + 4(-2) - 3 = 4 - 8 - 3 = -7$.

10.

Answer: D

Explanation:

- D is false. The correct formula for interior angle is $\frac{(n-2) \times 180^\circ}{n}$.
- A, B, and C are true properties of regular polygons.

11.

Answer: A

Explanation:

- Ratio of similarity: $\frac{9}{2} = 4.5$ (shorter side corresponds to 2cm).
- Other sides: $4 \times 4.5 = 18$ cm, $6 \times 4.5 = 27$ cm.
- Note: The first triangle (2,4,6) violates the triangle inequality ($2 + 4 = 6$), but for similarity, the ratios are used.

12.

Answer: C

Explanation:

- Similar triangles: $\frac{\text{height}}{\text{shadow}}$ is constant.
- $\frac{1.6}{2} = \frac{h}{1.8} \rightarrow h = 1.6 \times \frac{1.8}{2} = 1.6 \times 0.9 = 1.44$ m.

13. Answer A

1. 1. For Value 12:

The bar for value 12 reaches up to a frequency of 2.

2. 2. For Value 13:

The bar for value 13 reaches up to a frequency of 1.

3. 3. For Value 14:

The bar for value 14 reaches up to a frequency of 1.

4. 4. For Value 15:

The bar for value 15 has a frequency of 0 (no bar present).

5. 5. For Value 16:

The bar for value 16 reaches up to a frequency of 2.

6. **6. For Value 17:**

The bar for value 17 reaches up to a frequency of 1.

Now, let's compare these frequencies with the given tables:

- **Table A:**

V (Values) and f (Frequency) match the histogram's frequencies for each value: 12 (f=2), 13 (f=1), 14 (f=1), 15 (f=0), 16 (f=2), 17 (f=1).

- **Table B:**

The frequency for value 15 is 1, which is incorrect.

- **Table C:**

The frequency for value 13 is 2, which is incorrect.

- **Table D:**

The frequency for value 15 is 2, which is incorrect.
Therefore, Table A correctly represents the histogram.

The final answer is A.

14.

Answer: D

Explanation:

- Factors of 4: 1, 2, 4 (since $1 \times 4 = 4$, $2 \times 2 = 4$, $4 \times 1 = 4$).
- Outcomes: 3 favorable (faces 1, 2, 4) out of 6.
- Probability: $\frac{3}{6} = \frac{1}{2}$.

15.

Answer: D

Explanation:

- By vector addition, $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$.
- The figure (not shown) is standard for vector path addition.

16.

Answer: C

Explanation:

- Compute $f(x)g(x)$:
 $(x^3 + 2x + 1)(x^2 - 1) = x^3(x^2 - 1) + 2x(x^2 - 1) + 1(x^2 - 1) = x^5 - x^3 + 2x^3 - 2x + x^2 - 1 = x^5 + x^3 + x^2 - 2x - 1.$
- A: $f(x) + g(x) = x^3 + 2x + 1 + x^2 - 1 = x^3 + x^2 + 2x$ (not $x^3 + x^2$).
- B: $f(x) - g(x) = x^3 + 2x + 1 - (x^2 - 1) = x^3 + 2x + 1 - x^2 + 1 = x^3 - x^2 + 2x + 2$ (not $x^3 - x^2 + 2x$).
- D: $\frac{f(x)}{g(x)} = \frac{x^3 + 2x + 1}{x^2 - 1}$ (not simplified as given).

17.

D. If $f(a)$ and $f(b)$ have opposite signs, then there is at least one zero of f between a and b .

Explanation:

The Location Theorem (a consequence of the Intermediate Value Theorem) states that if a continuous function f changes sign over an interval $[a, b]$, then there must be at least one zero of f in that interval. Since polynomial functions are continuous, the theorem applies here. Option D correctly captures this idea by stating that opposite signs of $f(a)$ and $f(b)$ guarantee at least one zero between a and b .

18.

C. 1.69

Explanation:

To solve $5^x = 11.2$, take the logarithm (base 10) of both sides:

$$\log(5^x) = \log(11.2)$$

Using the logarithm power rule $\log(a^b) = b \log(a)$:

$$x \log(5) = \log(11.2)$$

Given $\log(5) = 0.6990$, and $\log(11.2)$ can be computed as:

$$\log(11.2) = \log(10 \times 1.12) = \log(10) + \log(1.12) = 1 + 0.0485 = 1.0485$$

Now solve for x :

$$x = \frac{\log(11.2)}{\log(5)} = \frac{1.0485}{0.6990} \approx 1.5$$

However, there seems to be a discrepancy with the provided options. Rechecking the calculation:

$$x = \frac{1.0485}{0.6990} \approx 1.5$$

But none of the options match exactly. Given the closest option is **C. 1.69**, it might be a typo or require re-evaluation. Alternatively, if the equation was $5^x = 11.2$ and the logarithm values are approximate, the closest option is **C**.

(Note: The calculation yields $x \approx 1.5$, but if the options are fixed, **C** is likely the intended answer based on the given logarithm values.)

19.

Explanation:

To find $\sqrt{48}$, express 48 as $48 = 16 \times 3$, so:

$$\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3} \approx 4 \times 1.732 = 6.928$$

Rounding to two decimal places, $\sqrt{48} \approx 6.93$.

Alternatively, using the given logarithm values:

$$\log(\sqrt{48}) = \log(48^{1/2}) = \frac{1}{2} \log(48)$$

But since $\log(4.8) = 0.6812$, we can approximate $\log(48)$ as:

$$\log(48) = \log(4.8 \times 10) = \log(4.8) + \log(10) = 0.6812 + 1 = 1.6812$$

Thus:

$$\log(\sqrt{48}) = \frac{1.6812}{2} = 0.8406$$

From the given values, $\log(6.93) = 0.8406$, so $\sqrt{48} \approx 6.93$.

20.

Explanation:

Combine the logarithms:

$$\log x + \log(x - 3) = \log[x(x - 3)] = 1$$

Exponentiate both sides (base 10):

$$x(x - 3) = 10^1 = 10$$

Solve the quadratic equation:

$$x^2 - 3x - 10 = 0 \implies (x - 5)(x + 2) = 0$$

Solutions: $x = 5$ or $x = -2$.

However, the domain of $\log x$ and $\log(x - 3)$ requires $x > 0$ and $x > 3$, so $x = 5$ is the only valid solution.

21.

Explanation:

The growth is exponential, modeled by:

$$N(t) = N_0(1 + r)^t$$

Given $N_0 = 1000$, $r = 5\% = 0.05$, and $t = 3$:

$$N(3) = 1000(1.05)^3 \approx 1000 \times 1.1576 = 1157.6$$

Rounding to the nearest whole number, $N(3) \approx 1158$.

22.

Answer: D. (3, 1)

Explanation:

Using the section formula for internal division in the ratio $m : n = 2 : 1$:

$$x = \frac{mx_2 + nx_1}{m + n} = \frac{2 \cdot 4 + 1 \cdot 1}{2 + 1} = \frac{8 + 1}{3} = 3$$
$$y = \frac{my_2 + ny_1}{m + n} = \frac{2 \cdot 2 + 1 \cdot (-1)}{3} = \frac{4 - 1}{3} = 1$$

Thus, the point is (3, 1).

23.

Answer: D. $2y - x - 12 = 0$

Explanation:

Using the point-slope form:

$$y - 5 = \frac{1}{2}(x - (-2)) \implies y - 5 = \frac{1}{2}(x + 2)$$

Multiply by 2:

$$2y - 10 = x + 2 \implies 2y - x - 12 = 0$$

24.

Answer: C. $\frac{x}{r}$

Explanation:

By definition, $\cos \theta = \frac{x}{r}$ (adjacent over hypotenuse in the right triangle formed).

25. Answer A

26.

Answer: C. $\csc 45^\circ = \sqrt{2}$

Explanation:

- $\csc \theta = \frac{1}{\sin \theta}$
- $\sin 45^\circ = \frac{\sqrt{2}}{2} \implies \csc 45^\circ = \frac{2}{\sqrt{2}} = \sqrt{2}$
- $\sin 135^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2} \implies \csc 135^\circ = \sqrt{2}$ (positive in Quadrant II).

27. Answer C

28.

Answer: B. 36π

Explanation:

Total surface area = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$:

$$2\pi \cdot 2 \cdot (7 + 2) = 4\pi \cdot 9 = 36\pi$$

29.

Answer: C. $36\sqrt{2}$

Explanation:

- Equilateral lateral faces imply slant height equals base side (6 cm).
- Height h from apex to base center: diagonal of base = $6\sqrt{2}$, half is $3\sqrt{2}$.

$$h = \sqrt{6^2 - (3\sqrt{2})^2} = \sqrt{36 - 18} = \sqrt{18} = 3\sqrt{2}$$

- Volume = $\frac{1}{3} \times \text{base area} \times h = \frac{1}{3} \times 6^2 \times 3\sqrt{2} = 36\sqrt{2}$.

30.

Answer: A. Part between base and parallel cross-section.

Explanation:

A frustum is the portion of a cone between the base and a plane parallel to the base, not passing through the vertex.

31.

Answer: B. $f(x) = (\sqrt{3})^x$

Explanation:

- Power functions: $f(x) = kx^a$ (e.g., $f(x) = \sqrt{2}|x|$ for A, $f(x) = x^5$ for C).
- $f(x) = (\sqrt{3})^x = 3^{x/2}$ is exponential, not power.

32.

Answer: B. Integer less than or equal to x .

Explanation:

The greatest integer function (floor function) returns the greatest integer $\leq x$.

33.

Answer: A. $\frac{1}{3-3x} + \frac{7}{3x+6}$

Explanation:

- Factor denominator: $x^2 + x - 2 = (x + 2)(x - 1)$.
- Decompose: $\frac{2x-3}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$.
- Solve: $2x - 3 = A(x - 1) + B(x + 2)$.
- Set $x = 1$: $-1 = 3B \implies B = -\frac{1}{3}$.
- Set $x = -2$: $-7 = -3A \implies A = \frac{7}{3}$.
- Result: $\frac{7/3}{x+2} - \frac{1/3}{x-1} = \frac{7}{3(x+2)} + \frac{1}{3(1-x)} = \frac{7}{3x+6} + \frac{1}{3-3x}$.

34. Answer A

35.

Answer: B. 5

Explanation:

Equate elements:

- $x + y = 7$
 - $x - y = 3$ (from bottom-right element).
- Add equations: $2x = 10 \implies x = 5$.

36.

Answer: B. $\begin{cases} x + 3y = 2 \\ 2x - 5y = 1 \end{cases}$ is non-homogeneous.

Explanation:

A homogeneous system has all constant terms zero. B has non-zero constants (2 and 1), so it is non-homogeneous.

37.

Answer: A. 0

Explanation:

Perform row operations:

- Row2 - 4 × Row1: $[2 - 4 \cdot 0.5, 2.5 - 4 \cdot 1, 3 - 4 \cdot 1.5] = [0, -1.5, -3]$
- Row3 - 7 × Row1: $[3.5 - 3.5, 4 - 7, 4.5 - 10.5] = [0, -3, -6]$
- Row3 - 2 × Row2: $[0, -3 - 2(-1.5), -6 - 2(-3)] = [0, 0, 0]$

Since a row is all zeros, determinant is 0.

38.

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Answer: B. $\begin{pmatrix} -1 & 5 & -1 \\ 0 & -3 & 1 \\ 1 & -4 & 1 \end{pmatrix}$

Explanation:

Multiply the matrix by option B:

- Result is $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, confirming it is the inverse.

39.

Answer: A. Scalars: magnitude only; vectors: magnitude and direction.

Explanation:

Standard definition: Scalars (e.g., mass) have magnitude; vectors (e.g., force) have magnitude and direction.

40.

Answer: D. Age: quantitative.

Explanation:

- Quantitative data is numerical (height, weight, age).
- Qualitative data is categorical (gender).
- Age is quantitative, so D is correct.

41.

Answer: A. 34.5

Explanation:

- The fourth class is 30–39.
- Midpoint = $\frac{\text{Lower Limit} + \text{Upper Limit}}{2} = \frac{30 + 39}{2} = \frac{69}{2} = 34.5$.

42.

- 2–4: $\frac{2+4}{2} = 3$
- 5–7: $\frac{5+7}{2} = 6$
- 8–10: $\frac{8+10}{2} = 9$
- 11–13: $\frac{11+13}{2} = 12$
- Sum of (midpoint × frequency):
 - $3 \times 4 = 12$
 - $6 \times 1 = 6$
 - $9 \times 2 = 18$
 - $12 \times 3 = 36$
 - Total = $12 + 6 + 18 + 36 = 72$
- Total frequency = $4 + 1 + 2 + 3 = 10$
- Mean = $\frac{72}{10} = 7.2$.

43.

Answer: B. 8, 7, 11, 13, 17, 9, 8, 7, 6, 11

Explanation:

- Multimodal means having more than one mode (value with highest frequency).
- A: Sorted: 4, 6, 6, 9, 10, 10. Modes: 6 and 10 (bimodal).
- B: Sorted: 6, 7, 7, 8, 8, 9, 11, 11, 13, 17. Modes: 7, 8, 11 (each appears twice, trimodal).
- C: All values unique (no mode).
- D: Sorted: 3, 10, 11, 12, 14, 15, 18, 18. Mode: 18 (unimodal).
- B has the most modes (trimodal).

44.

Answer: B. Permutations: 24

Explanation:

- "Chosen 3 at a time" implies selection, but the question asks for the true statement about permutations/combinations.
- Combinations (order irrelevant): $C(4, 3) = \binom{4}{3} = 4$.
- Permutations (order relevant): $P(4, 3) = 4 \times 3 \times 2 = 24$.
- Only B is correct.

45.

Answer: D. Outcomes that ensure the occurrence of a particular result.

Explanation:

- Favorable outcomes are those that satisfy the condition of a specific event.
- D correctly describes this.

46.

Answer: D. Ratio between consecutive terms is a non-zero constant.

Explanation:

- A geometric progression has a constant ratio between consecutive terms.
- The ratio can be negative (e.g., 2, -4, 8, -16), so it must be non-zero but not necessarily positive.

47.

Answer: D. 3069

Explanation:

- Geometric sequence: first term $a = 3$, common ratio $r = 2$.
- Partial sum of first n terms: $S_n = a \frac{r^n - 1}{r - 1}$.
- $S_{10} = 3 \frac{2^{10} - 1}{2 - 1} = 3 \times (1024 - 1) = 3 \times 1023 = 3069$.

48.

Answer: B. $\sum_{n=1}^{\infty} 2^n$ divergent.

Explanation:

- A geometric series $\sum ar^n$ converges if $|r| < 1$.
- A: $r = \frac{4}{3} > 1$, diverges.
- B: $r = 2 > 1$, diverges (true).
- C: $r = -\frac{2}{5}$, $|r| = \frac{2}{5} < 1$, converges.
- D: $r = -2$, $|r| = 2 > 1$, diverges.

49.

Answer: A. 41

Explanation:

- Average rate of change = $\frac{g(b) - g(a)}{b - a}$ for $[a, b]$.
- $g(6) = 6^3 - 2 \times 6 = 216 - 12 = 204$.
- $g(1) = 1^3 - 2 \times 1 = -1$.
- $\frac{204 - (-1)}{6 - 1} = \frac{205}{5} = 41$.

50.

Answer: C. $\frac{3}{2}x^{-\frac{1}{2}} + \frac{4}{3}x^3 + 2$

Explanation:

- The options are derivatives, so compute $(f + g)'(x)$.
- $f(x) = 3x^{1/2} \implies f'(x) = 3 \times \frac{1}{2}x^{-1/2} = \frac{3}{2}x^{-1/2}$.
- $g(x) = \frac{1}{3}x^4 + 2x \implies g'(x) = \frac{4}{3}x^3 + 2$.
- $(f + g)'(x) = f'(x) + g'(x) = \frac{3}{2}x^{-1/2} + \frac{4}{3}x^3 + 2$.

51.

Answer: C. $-\frac{1}{\sqrt{1-2x}}$

Explanation:

- $f(x) = (1 - 2x)^{1/2}$.
- $f'(x) = \frac{1}{2}(1 - 2x)^{-1/2} \times (-2) = -\frac{1}{\sqrt{1-2x}}$.

52.

Answer: C. (2,-1) and (0,3)

Explanation:

- Critical points: $f'(x) = 3x^2 - 6x = 3x(x - 2) = 0 \implies x = 0, 2$.
- Second derivative: $f''(x) = 6x - 6$.
- $f''(0) = -6 < 0 \rightarrow$ local maximum at $x = 0$, $f(0) = 3 \rightarrow$ point (0,3).
- $f''(2) = 6 > 0 \rightarrow$ local minimum at $x = 2$, $f(2) = 8 - 12 + 3 = -1 \rightarrow$ point (2,-1).

53.

Answer: B. The degree to which numerical data tends to distribute about an average.

Explanation:

- Dispersion quantifies how data spreads around a central value (average).
- B is the standard definition.

54.

Answer: A. 2.75

Explanation:

- Mode = 19 (highest frequency, 4 times).
- Deviations from 19: $|13 - 19| = 6$, $|16 - 19| = 3$, $|19 - 19| = 0$ ($\times 4$), $|24 - 19| = 5$, $|27 - 19| = 8$.
- Sum of deviations = $6 + 3 + 0 + 0 + 0 + 0 + 5 + 8 = 22$.
- Number of data points = 8.
- Mean deviation = $\frac{22}{8} = 2.75$.

55.

Answer: A. 0.35

Explanation:

- Sorted data: 10, 15, 25, 29, 35, 35, 44, 46, 52, 55, 56 ($n = 11$).
- Q_1 position: $\frac{n+1}{4} = 3\text{rd term} \rightarrow 25$.
- Q_3 position: $\frac{3(n+1)}{4} = 9\text{th term} \rightarrow 52$.
- Quartile deviation coefficient = $\frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{52 - 25}{52 + 25} = \frac{27}{77} \approx 0.35$.

56.

Answer: B. $\frac{5}{8}$

Explanation:

- Sorted data: 1, 3, 7, 9, 11, 13, 18, 28 ($n = 8$, even).
- Median = $\frac{9+11}{2} = 10$.
- Deviations from median: $|1 - 10| = 9$, $|3 - 10| = 7$, $|7 - 10| = 3$, $|9 - 10| = 1$, $|11 - 10| = 1$, $|13 - 10| = 3$, $|18 - 10| = 8$, $|28 - 10| = 18$.
- Sum of deviations = $9 + 7 + 3 + 1 + 1 + 3 + 8 + 18 = 50$.
- Mean deviation = $\frac{50}{8} = 6.25$.
- Coefficient = $\frac{\text{Mean deviation}}{\text{Median}} = \frac{6.25}{10} = 0.625 = \frac{5}{8}$.

57.

Answer: B. -2 and -6

Explanation:

- Feasible region vertices:
 - $(4,1)$: $x + y = 5$ and $x + 2y = 6 \rightarrow Z = -4 + 2(1) = -2$.
 - $(5,0)$: $x + y = 5$ and $y = 0 \rightarrow Z = -5 + 0 = -5$.
 - $(6,0)$: $x + 2y = 6$ and $y = 0 \rightarrow Z = -6 + 0 = -6$.
- Maximum $Z = -2$ at $(4,1)$, minimum $Z = -6$ at $(6,0)$.

58.

Answer: A.

Explanation:

- Cost per flat screen: 15,000 Birr, curved: 20,000 Birr.
- Total investment limit: 900,000 Birr $\rightarrow 15,000x + 20,000y \leq 900,000$.
- Storage capacity: 50 TVs $\rightarrow x + y \leq 50$.

59.

Answer: B. 132

Explanation:

- Let regular price be P .
- After 25% discount: $P - 0.25P = 0.75P = 99$.
- $P = \frac{99}{0.75} = 132$.

60.

Answer: A. 118,000

Explanation:

- Down payment: $10\% \text{ of } 100,000 = 10,000$.
- Monthly installments: $36 \times 3,000 = 108,000$.
- "Plus interest" is ambiguous, but the total installment amount (including interest) is Birr 3,000 per month.
- Total paid = Down payment + Installments = $10,000 + 108,000 = 118,000$.

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